AIM :

To write a Python program to implement Breadth First Search (BFS) traversal.

ALGORITHM:

1. Start the program.
2. Create a graph using a dictionary.
3. Initialize a queue with the starting vertex.
4. Remove the first vertex from the queue.
5. Visit the vertex if it is not already visited.
6. Add its unvisited neighbours to the queue.
7. Repeat until the queue becomes empty.
8. Stop the program.

PROGRAM :

prgm7.py - C:/Users/hivet/AppData/Local/Programs/Python/Python314/prgm7.py (3.14.0)

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```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
        vertex = queue.popleft()

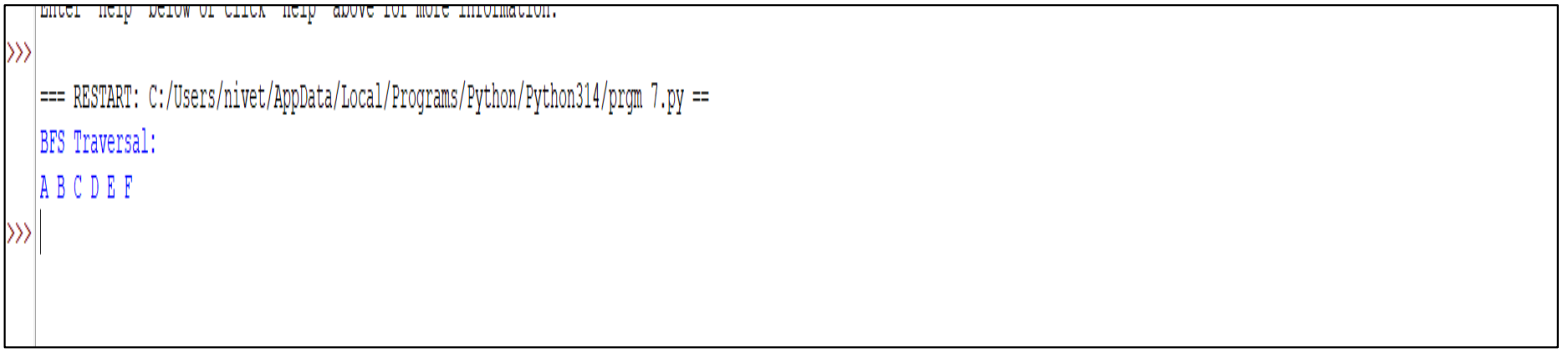
        if vertex not in visited:
            print(vertex, end=" ")
            visited.add(vertex)

            for neighbour in graph[vertex]:
                if neighbour not in visited:
                    queue.append(neighbour)

# Graph representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("BFS Traversal:")
bfs(graph, 'A')
```

OUTPUT :



RESULT :

Thus, the Python program to implement BFS traversal was executed successfully.